

Features of Qucs

Let us take a look at the main features of Qucs:

Massive component library. Qucs comes up with almost all the components required for an electronic circuit. The library consists of lumped elements like resistors, inductors

and capacitors as well as models for sources, probes and transmission lines for electronic circuits.

We can also find non-linear components like diodes and transistors in the library. It includes various transistor models like heterojunction bipolar transistor (HBT), metal semiconductor field effect transistor (MESFET), metal oxide semiconductor field effect transistor (MOSFET) and junction gate field effect transistor (JFET) in the component library. Additionally, users of this tool are given the provision for expanding the libraries as per their requirements.

Smart GUI. You have access to a very useful GUI in Qucs that is based on Qt by Digia. This helps us create schematics or write VHDL code using the user interface. Setting up simulations and displaying their results is very easy using this GUI.

The GUI aids in efficient schematic capture by allowing the simulated data to be represented in various forms as Smith chart, tabular, cartesian, polar, Smith-polar combination, 3D-cartesian, locus curve, timing diagram and truth table.

Analysis of results made easy. Various types of analyses are possible with the help of Qucs. We can get the small signal response of a circuit using AC analysis, while the DC sweep analysis in Qucs could be used to calculate the circuit's bias point over a range of values. The transient analysis helps us compute the response of the circuit with respect to time.

Not only can we easily calculate the contribution of noise from the resistor and semiconductor devices in the circuit with the help of noise analysis, we can also vary the range of values of a component in the circuit and note the response using Qucs with the help of parameter sweep analysis. It is easy to obtain scattering parameters by using S parameter simulation in Qucs, while pure digital simulations are supported by VHDL and Verilog HDL in Qucs.

What users think about this tool

The strength of any tool lies in the experience that it can provide its users. Going through the reviews in the online communities, we realised that

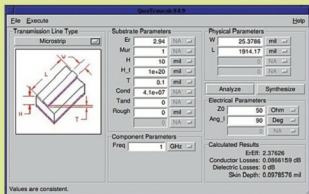


Fig. 2: Transmission line calculator

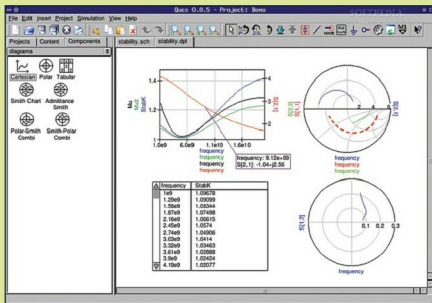


Fig. 3: Plotting data in Qucs

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